

Means of Reducing Ringing Times in Pulsed Nuclear Magnetic Resonance

John J. Spokas

Citation: [Review of Scientific Instruments](#) **36**, 1436 (1965); doi: 10.1063/1.1719351

View online: <http://dx.doi.org/10.1063/1.1719351>

View Table of Contents: <http://scitation.aip.org/content/aip/journal/rsi/36/10?ver=pdfcov>

Published by the [AIP Publishing](#)

Articles you may be interested in

[Study of chemical gelation dynamics of acrylamide in water by realtime pulsed nuclear magnetic resonance measurement](#)

J. Chem. Phys. **89**, 3363 (1988); 10.1063/1.454945

[Realtime measurement of crystallization process of polymers by pulsed nuclear magnetic resonance](#)

J. Appl. Phys. **59**, 1488 (1986); 10.1063/1.336504

[Powerful timing generator using monochip timers: An application to pulsed nuclear magnetic resonance](#)

Rev. Sci. Instrum. **53**, 1 (1982); 10.1063/1.1136800

[Acoustic ringing effects in pulsed nuclear magnetic resonance probes](#)

Rev. Sci. Instrum. **49**, 1151 (1978); 10.1063/1.1135538

[Periodically Pulsed Nuclear Magnetic Resonance](#)

J. Chem. Phys. **43**, 76 (1965); 10.1063/1.1696491

Nor-Cal Products



Manufacturers of High Vacuum
Components Since 1962

- Chambers
- Viewports
- Valves
- Motion Transfer
- Foreline Traps

- Flanges & Fittings
- Feedthroughs



www.n-c.com
800-824-4166

Means of Reducing Ringing Times in Pulsed Nuclear Magnetic Resonance*

JOHN J. SPOKAS†

Argonne National Laboratory, Argonne, Illinois

(Received 26 April 1965; and in final form, 18 June 1965)

Two auxiliary circuits are described for incorporation in a pulsed nuclear magnetic resonance system to reduce the deadtime of the system following an rf exciting pulse. The function of one, a driven balanced clamp, is to reduce the Q of the detecting circuit for a controlled time after each pulse. The other, a symmetric damping circuit requiring little operating power, works in conjunction with the transmitter to "square-up" the rf pulses generated. The circuits presented here represent certain improvements in established techniques of clamping and damping. Additionally, it is pointed out that there is a particular advantage, with respect to minimizing the deadtime in a coherent system, of using timing pulses which are phase locked to the rf reference.

INTRODUCTION

IN pulsed nuclear resonance experiments one is confronted with the task of momentarily (i.e., for a period of 1–10 μ sec duration) impressing a radio frequency magnetic field of several hundred gauss amplitude ($2H_1$) on a sample of, let us say, 1 cm^3 volume and of detecting the weak nuclear induction signal in the same sample immediately following the removal of the rf field. There are two common approaches to this problem. In one approach, two separate orthogonal coils are placed around the sample, one for impressing the intense rf magnetic field on the sample, the second for detecting the nuclear induction signal. In the other approach¹ again two coils are used but only one surrounds the sample. The two coils are connected in a bridge circuit wherein the difference in voltage across the coils is detected. At balance the driving voltages across the coils are equal and the intense driving voltage is not detected but the nuclear induction signal, being generated in only one coil, is detected.

In principle a perfectly orthogonal coil combination or a perfectly symmetric bridge would allow the induction signal to be observed immediately after the driving pulse. But, alas, the degree of perfection one can practically attain and maintain still results in a signal voltage into the rf amplifier which, depending on the Q of the detecting circuit and the characteristics of the amplifier, may block the amplifier for several tens of microseconds. The two auxiliary circuits described herein allow this limitation to be overcome to a degree greater than before and are applicable equally well to the crossed-coil or bridge methods.

The present clamp allows a higher Q , hence more sensitive, circuit to be employed while minimizing the ringing transient from the radio frequency pulse which causes the nuclear induction signal. The damping circuit operates on the output of the pulse transmitter in such a way as to sharply, yet efficiently, reduce the rise- and fall-time of the radio frequency pulse. Such circuits have been described

previously.^{1,2} The circuits described here represent refinements of some of the previous schemes.

THE BALANCED CLAMP

In clamping circuits previously described, a semiconductor diode is connected across the tuned detecting LC circuit. The diode is normally zero biased in which circumstance its resistance is very large and so it does not affect the detecting circuit. The usual circuit is illustrated in Fig. 1(a).

A square voltage pulse impressed across R holds the diode forward biased for the entire pulse duration. Thus, during the pulse, the resistance R in shunt with the output impedance of the pulse generator, which combined is kept under 100 Ω or so, is connected across the tuned circuit. The circuit Q is drastically lowered in this state and the decay of spurious excitations caused by the transmitter pulse is correspondingly sped up. The time this low Q condition is maintained is adjusted empirically to minimize the recovery time of the system.

A limitation of this arrangement arises from the fact

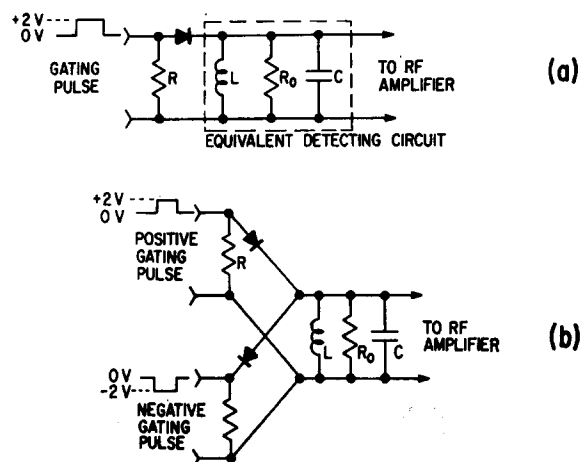


FIG. 1. Illustration of single-ended and double-ended clamp circuits. (a) The customary single-ended arrangement. (b) A balanced two diode arrangement where the forward bias current does not pass through the tuned circuit inductance L .

² W. Gilbert Clark, *Rev. Sci. Instr.* 35, 316 (1964).

* This work performed under the auspices of the U. S. Atomic Energy Commission.

† Permanent address: St. Procopius College, Lisle, Illinois.

¹ Irving J. Lowe and Dennis E. Barnaal, *Rev. Sci. Instr.* 34, 143 (1963).

that the forward bias current flows through the coil and cannot be abruptly reduced to zero without itself producing a ringing transient in the detecting circuit. Lowe¹ has described how this transient can be reduced by shaping the trailing edge of the gating pulse.

The clamp circuit described here uses two diodes in a balanced arrangement wherein the forward current of one diode passes through the other diode and ideally none through the coil. This allows the low Q state to be suddenly changed to a high Q state and yet not cause a ringing transient. Such a circuit is illustrated in Fig. 1(b). Meiboom and Gill³ use a balanced clamp requiring a single-ended gating pulse. Their circuit employs a bridge-type detecting circuit to cancel the ringing transients caused by the gating pulse.

The double gating pulse required by the balanced clamp described here is conveniently obtained by local rectification of a rf gating pulse. The actual circuit that has been in use for the past year is shown in Fig. 2. The clamp circuit is situated on the rf amplifier chassis and as near to the input grid as possible. The frequency of the gating pulse is near 35 Mc which is sufficiently removed from the center frequency of the amplifier to not be picked up directly by the amplifier. The 50 Ω potentiometer enables the rectified pulses to be equalized in amplitude and the trimmer capacitor permits the shapes of the trailing edges of the rectified pulses to be equalized. The performance of this circuit is illustrated in Fig. 3.

A feeble ringing transient is still present which is undoubtedly due to imperfect symmetry of the clamp. The bad effects of this transient can be minimized in a coherent pulsed NMR system employing phase sensitive detection of the nuclear signals. This further improvement, however, requires that the timing pulses be phase-locked to the rf reference.⁴ In this circumstance a slight alteration in the length of the clamp gating pulse will shift the position of its trailing edge in the rf cycle. This, in essence, affords a control of the phase of the ringing transient. By thereby setting the phase of the ringing transient to be in quad-

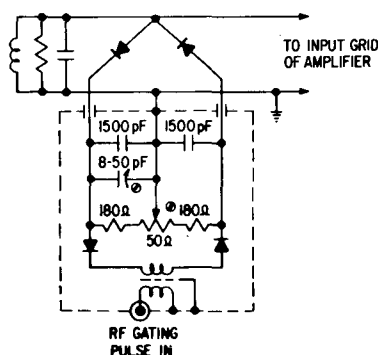


FIG. 2. A balanced clamp circuit which derives the double-ended gating pulse from rectification of a rf pulse. All diodes are type SG-5000 manufactured by Transitron.

³ S. Meiboom and D. Gill, Rev. Sci. Instr. 29, 688 (1958).

⁴ A timing pulse generator designed specifically for pulsed NMR work and having this special feature has been in use for the past year and will be described in a future publication.

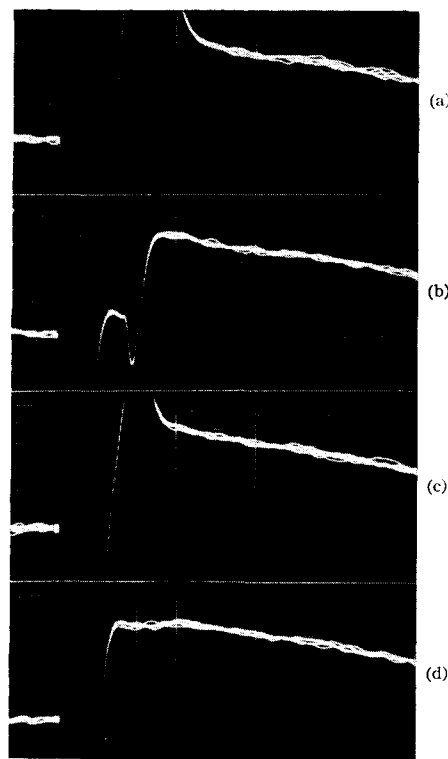


FIG. 3. Oscilloscope displays of the detector output illustrating the effectiveness of the clamp shown in Fig. 2. (a) The initial portion of the Bloch decay in aluminum following a 90° pulse. No damping in transmitter and clamp not operating. (b) Same as (a) except clamp is operating. (c) Same as (a) except there is damping of the transmitter output. (d) Same as (a) except there is damping of the transmitter and clamp is operating. Sweep speed is 5 $\mu\text{sec}/\text{div}$, radio frequency is 8 Mc, and $H_1 = 110$ G.

rature with the nuclear induction signal, the ringing transient is effectively discriminated against and will not appear at the output of the phase-sensitive detector. It is assumed that the phases of the rf reference and signal are either equal or 180° apart which is the normal mode of operation. Clearly the advantage of phase control of the ringing transient would also be realized in conjunction with other types of gated clamping circuits.

THE BALANCED DAMPING CIRCUIT

The utility of a circuit to rapidly quench the output of the rf transmitter in pulsed NMR experiments has already been pointed out. The effectiveness of a clamp circuit such as described above is only realized after the rf output of the transmitter has decayed to a level which is 10^6 or so lower than its level during the pulse. This requires 20 times the characteristic time of the circuit being driven by the transmitter. Hence, quick receiver recovery by itself will not lead to the shortest deadtime for the system; rapid decay of the excitation is also necessary. In the "brute force" approach to expedite the falltime of the transmitter bursts, one would merely lower the Q of the driven circuit by resistive loading thus reducing the re-

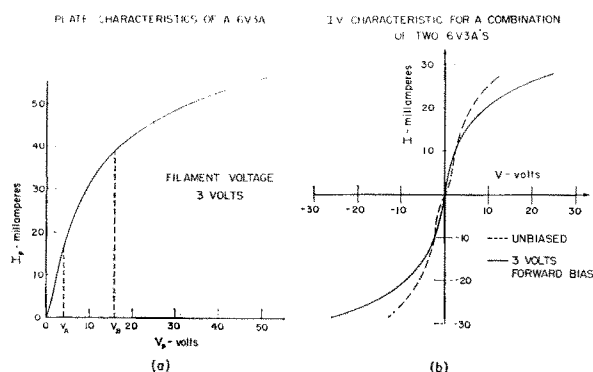


FIG. 4. Typical plate characteristics of a 6V3A thermionic diode operated at reduced filament voltage. (a) Single-diode characteristic. (b) Symmetric IV characteristic of a parallel combination of two diodes, one being inverted relative to the other.

covery time to an arbitrary small value. However, the costs in terms of rf pulse power can become excessive. Using a nonlinear resistance in parallel with the driven circuit, as Clark² does, allows this constraint to be violated appreciably.

A symmetric damping circuit which shows excellent characteristics while requiring relatively little dc power to operate has proven itself in use over the past year. The advantage of low power consumption is derived from the fact that that nonlinearity is used which appears at high voltage across the diode when the diode current becomes limited by emission from the cathode. All previous damping circuits have used the sharp nonlinearity which occurs when the voltage across the diode goes through zero, the plate to cathode resistance becoming nearly infinite in the reverse direction.

A typical plate characteristic of a thermionic diode is shown in Fig. 4(a). Up to the plate voltage V_A , the curve follows Child's law with the current increasing as the $\frac{3}{2}$ power of the voltage. For voltages above V_B , the curve flattens out appreciably as the current becomes limited by emission. The dynamic impedance dV/dI is seen to be relatively large at voltages near zero. It decreases to a minimum in the neighborhood of V_A . As the voltage is raised beyond V_A the impedance increases rapidly becoming very large beyond V_B . The shape of the curve above V_A then exhibits just the feature sought for damping, namely, an impedance which is low but becomes very large at high voltages.

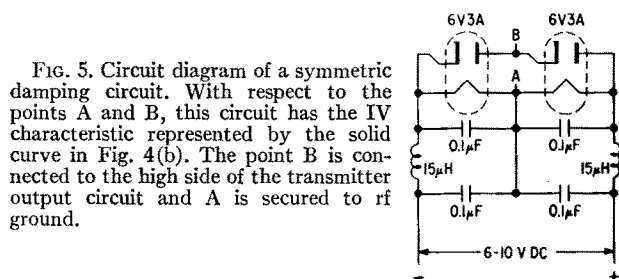


FIG. 5. Circuit diagram of a symmetric damping circuit. With respect to the points A and B, this circuit has the IV characteristic represented by the solid curve in Fig. 4(b). The point B is connected to the high side of the transmitter output circuit and A is secured to rf ground.

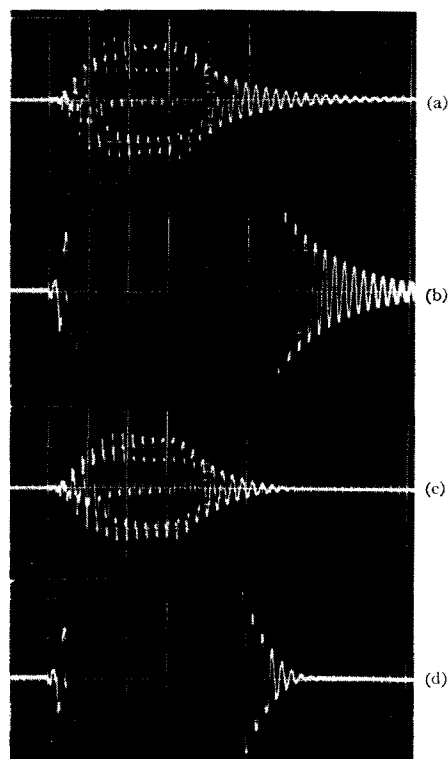


FIG. 6. Oscilloscope presentation of transmitter output with and without damping. (a) Transmitter output corresponding to (a) and (b) of Fig. 3, no damping (circuit of Fig. 5 is connected but filament voltage is 0). Vertical sensitivity is 2800 V/div. (b) Identical to (a) except vertical sensitivity is increased tenfold. (c) and (d) Damping obtained under conditions of (a) and (b), respectively, with 4.0 V across the filament of each 6V3A. Sweep speed is 0.5 μ sec/div.

Actually with respect to the present application of radio frequency damping, it is the impedance averaged over an entire period that matters. But this will exhibit the same variation with amplitude of the radio frequency voltage, though blunted to an extent, as the dc dynamic impedance discussed above. The voltage at which the transition between the low and high impedance states occurs can be identified as V_B in Fig. 4. This transition voltage can be varied over a limited range through the cathode temperature which, in turn, is governed by the power supplied to the filaments. The effect of the damping circuit is to present a resistance which decreases with decreasing rf amplitude, so that, as the free oscillations decay, the time constant becomes progressively shorter.

Using two like diodes in parallel with the anode of one joined to the cathode of the other yields an over-all impedance curve which is highly symmetric [see Fig. 4(b)]. More significant, however, is the relatively little dc biasing power the proposed arrangement requires by comparison to Clark's two diode arrangement.² It is definitely advantageous to forward bias both diodes in order to avoid the high impedance region near zero plate voltage that is evident in the dotted curve of Fig. 4(b). However, 3–5 V is quite sufficient.

The actual circuit that has been in use for several months is shown in Fig. 5. Direct current from a "floating" supply whose output can be varied from 6–10 V is used both to heat the filaments and to provide a small forward bias to each diode. The supply is grounded at the damping circuit as shown. Besides requiring just one supply, this also allows neat wiring of the damping circuit. At 7 V across the series combination of two filaments, which allows 3.5 V forward bias on each diode, the diode plate current is 15 mA. Point A is connected to rf ground which is also dc ground. Point B is joined to the high side of the circuit to be damped.

This circuit has been in use for several months to damp rf pulses in excess of 4000 V peak amplitude. No degradation in performance has been noted during this time. The circuit has not been tested at voltages much greater than 5000 V. Arcing within the 6V3A's will likely limit the maximum useful operating voltage to a value not much greater than 5000 V. The performance of this circuit is shown in Fig. 6.

It is observed that a single diode having the character-

istic curve shown in Fig. 4(a) and biased at a suitable value between V_A and V_B exhibits a somewhat symmetrical over-all impedance variation about the bias point. That is, the impedance is low for small positive or negative voltages and large for large positive or negative voltages. Such an arrangement has been demonstrated using a single 6V3A operated at the rated filament power and gives entirely satisfactory damping. The chief disadvantage to such a single-diode circuit is the greater dc power required to maintain the greater bias and the lack of symmetry leading to greater harmonic generation in the rf pulses.

ACKNOWLEDGMENTS

The author is grateful to Argonne National Laboratory for summer appointments held during 1963 and 1964 which made this work possible. Thanks are extended to Dr. D. O. Van Ostenburg and Dr. D. J. Lam for their continued support and critical reading of the manuscript. John Brewer provided stimulation, made valuable suggestions concerning details of the actual circuits, and critically read the manuscript.

Vacuum Tight Windows with Wide Band Transmission Characteristics*

T. P. VOGL, R. O. MCINTOSH,[†] AND M. GARBUNY

Westinghouse Electric Corporation, Research & Development Center, Beulah Road, Churchill Borough, Pittsburgh, Pennsylvania 15235

(Received 10 June 1965; and in final form, 8 July 1965)

A vacuum tight barium fluoride window with transmission range from 1800 Å to 14 μ is described. The window consists of a barium fluoride disk sealed into a suitably formed fine silver spinning with silver chloride. Evacuated envelopes incorporating this window have been made and kept for months without indication of leakage, based on the performance of photoelectric cathodes in the vacuum space. This technique has been equally successful in sealing lithium fluoride windows. The sealing procedure is fully discussed.

INTRODUCTION

FOR many purposes a vacuum tight window capable of transmitting all radiation from the ultraviolet to the medium infrared is desirable, and such a window utilizing barium fluoride has been developed. The window assembly consists of a steel supported silver frame, a barium fluoride window, and silver chloride filler. The silver chloride is melted in a controlled atmosphere furnace with a pre-set temperature cycle to form a vacuum tight bead around the periphery of the window. Evacuated envelopes incorporating this window have been made and kept for months without indication of leakage.

* Work performed as part of U. S. Air Force Contract AF-33(616) 2282.

[†] Present Address: Bendix Research Laboratories, Detroit, Michigan.

PRINCIPLE OF THE SEAL¹

The principal capillary seal is formed in region A, Fig. 1, along the raised bead of the window frame, the only portion of the structure which is in continuous contact during the sealing cycle, independent of thermal expansion coefficients.

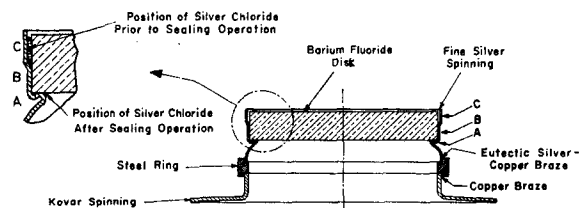


FIG. 1. Barium fluoride window assembly.

¹ U. S. Patent 2,966,592 assigned to the Westinghouse Electric Corporation.